

GaToroid – getting closer to affordable cancer therapy

CERN has been contributing to developing technologies to improve hadron therapy for the past 30 years. Now, a team at CERN is working on new design concepts that would make the machines delivering hadron therapy much more compact and economically viable



Luca Bottura, project lead of GaToroid, and Gregory Maury from the TE/MS-SMT section are working on winding a coil of the demonstrator for GaToroid. (Image: CERN)

Hadron therapy is a state-of-the-art radiotherapy technique that uses proton or ion beams to target tumoural cells, while sparing surrounding healthy tissues from unwanted radiation. To achieve best results, it requires complex systems, called gantries, that rotate around the patient, carrying magnets that guide the hadron beams precisely onto tumours. Unfortunately, these are usually massive and costly machines, which represent a limiting factor for the spreading of the treatment technique. Drawing on its expertise in superconducting magnets, a team at CERN has developed a novel superconducting and lightweight gantry, [GaToroid](#), which promises to be much more compact than current machines while being just as precise. Unlike traditional rotating gantries, GaToroid could deliver particle beams from different directions without requiring the patient or the magnets to be moved. It exploits compact [superconducting coils](#), located at different angles around the patient, to create a steady-state toroidal magnetic field that can direct particle beams onto a tumour, from any direction, sparing the patient from any stray magnetic field.

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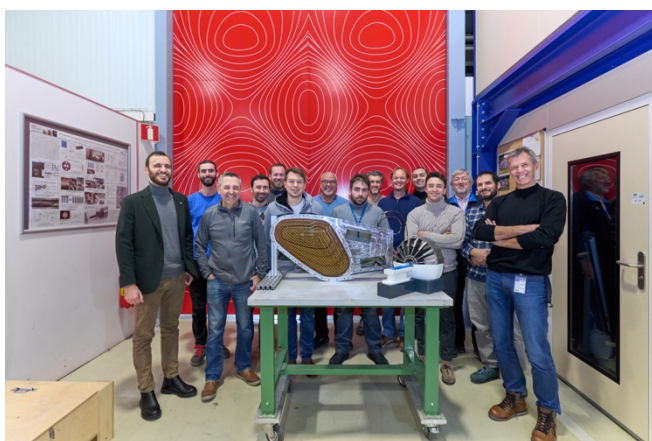
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Understanding and overcoming “resistance to change”

The use of superconductors gives the GaToroid design its unique compactness, with a diameter of five metres and an estimated weight of approximately 12 tonnes for proton beams. This represents a substantial weight reduction compared to traditional gantries for proton applications, which can weigh up to 270 tonnes. In 2023, the team constructed and tested a demonstrator magnet consisting of one coil for the proton version of GaToroid, scaled down to one-third of the real size. The goal of the demonstrator was to prove the mechanical resistance of the structure and to check the quality of the field. The superconductor of choice was niobium–titanium, the same material used for the cables of the LHC magnets. “The tests we ran with our GaToroid demonstrator were highly successful, showing excellent performance and demonstrating that this magnet technology is ready for this type of application,” says Luca Bottura, head of the project.



The team that has designed, built and tested the demonstrator coil for the GaToroid project, next to a scaled 1:3 version of it. From left to right: Enrico Felcini, Cedric Urscheler, Juan Carlos Perez, Francois-Olivier Pincot, Pierre-Antoine Contat, Melvin Liebsch, Ahmed Benfkih, Mariano Pentella, Gregory Maury, Sébastien Clement, Carlo Petrone, Gianluca Vernassa, Gijs de Rijk, Ariel Haziot, Luca Bottura. (Image: M. Cavazza/CERN)

The team performed several tests on the magnet and collected precise measurements of the

magnetic field profile. “The magnet generated a field that matched the expected one with an error of less than 1%, and it showed absolutely no signs of mechanical degradation following the powering tests,” explains Gianluca Vernassa, who performed the mechanical and electromagnetic calculations on the demonstrator magnet.

This innovative design has already garnered significant attention within the particle therapy community across Europe. Notably, [PARTREC](#) has expressed interest in collaborating with the team to explore how GaToroid could be adapted for [FLASH therapy](#).

Unlike in traditional radiotherapy, where treatment doses are delivered sequentially from different directions with long pauses to adjust the rotating gantry, FLASH radiotherapy delivers the entire dose from all directions simultaneously within 100 milliseconds. While conventional rotating gantries cannot operate in this mode due to their inherent limitations, the GaToroid system, with its fixed structure and ability to deliver beams from multiple angles without mechanical adjustments, would be perfectly suited to this type of treatment. This makes GaToroid a potential enabling technology for the use of hadrons in FLASH radiotherapy, which has not been possible with existing systems.

Following the success of the demonstrator, several GaToroid designs tailored to FLASH therapy have been proposed by the team. These include a conceptual design for very high-energy electrons (VHEE) and a proton-based system for which a proposal has already been developed in collaboration with PARTREC. The team is actively working on these developments and expects new results in 2025.

For more information about GaToroid consult the [Knowledge Transfer website](#).

Melanie Arnold

Accelerator Report: Preparing for 2025 LHC operation and setting goals

The Chamonix Workshop is in full swing, and the initial outlines of the 2025 LHC configuration and operational scenarios are beginning to take shape

The year-end technical stop (YETS) activities in the accelerator complex's first machines are nearing completion, but in the LHC about six weeks of work still lie ahead of us before the machine is handed over to the Operations group on 13 March for recommissioning.

While having a well-defined and approved schedule is essential, an equally critical element for the 2025 run is establishing clear operational configurations and scenarios. These are developed and refined throughout the year in various meetings, working groups and committees. They are then consolidated in two key workshops, which are vital not only to ensure the immediate operation of the accelerator complex but also to prepare its long-term future.

The first of these is the [Joint Accelerator Performance Workshop](#) (JAPW), which took place from 10 to 12 December 2024. The workshop brought together accelerator and operations teams from across the complex, along with teams from the experiments, to review the performance achievements, challenges and lessons from 2024. The focus was on defining the configuration and performance goals for the 2025 run.

The second, more strategic, workshop is the [Chamonix Workshop](#), which goes beyond operational aspects to address broader topics. These include planning for the remainder of Run 3 and for Run 4, as well as looking ahead to Long Shutdown 3 (LS3), the High-Luminosity LHC (HL-LHC) and the Future Circular Collider (FCC) Feasibility Study. Importantly, the Chamonix Workshop integrates input and recommendations from the JAPW and makes key decisions that will guide the operation of the accelerator complex in the years leading up to LS3.

As I write, the Chamonix Workshop is in full swing, and the initial outlines of the 2025 LHC configuration and operational scenarios are beginning to take shape.

For the 2025 LHC run, the majority of the schedule is dedicated to proton collisions, complemented by shorter runs for lead and oxygen ions. Based on

the approved 2025 LHC schedule and the beam parameters discussed, preliminary target figures have been proposed, pending final validation and, of course, assuming equal or even better beam availability throughout the whole accelerator complex than the excellent availability in 2024.

In 2024, the proton run was 147 days long and delivered an integrated luminosity of 124 fb^{-1} . For 2025, 138 days have been allocated for proton collisions – 9 days fewer than in 2024. Despite the shorter runtime, the proposed targets for integrated luminosity are ambitious: 120 fb^{-1} for ATLAS and CMS, 12 fb^{-1} for LHCb and 50 pb^{-1} for ALICE. Achieving these goals will be challenging, but realistic, thanks to the potential to slightly increase luminosity production by exploiting small improvements in electron cloud limitations (see box below).

For the 21-day lead-ion run at the end of 2025, the proposed integrated luminosity targets are 2.4 nb^{-1} for ATLAS, CMS and ALICE, and 0.8 nb^{-1} for LHCb. While these targets are also demanding, lessons learned from the successful 2024 lead-ion run could help achieve further optimisations, making these goals challenging but realistic.

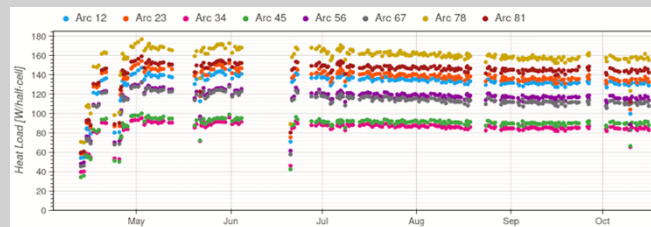
Rende Steerenberg

How can we slightly increase luminosity production?

The LHC beam consists of “bunch trains”, which are groups of closely spaced particle bunches, each of which is separated by 25 nanoseconds (about 30 cm) from the next one. These bunch trains vary in length, depending on the number of consecutive bunches. As the beam circulates through the LHC's vacuum chambers and beam screens (tubes that shield the magnets' coils and cryogenic system from the heat loads, radiation and other damage), it releases free electrons, which are accelerated by the passing bunches. These electrons then collide with the beam screen, releasing more electrons in a chain reaction known as “electron cloud” production.

This electron cloud can cause beam instabilities. Furthermore, the collisions of electrons with the beam screen generate heat, which must be removed by the cryogenic cooling system. The heat produced limits both the number of bunches in a train and the intensity of each bunch. Shorter bunch trains and larger gaps between them reduce the formation of electron clouds and, in turn, the heat deposited on the beam screen.

In 2024, the LHC operated with bunch trains composed of 3 batches of 36 bunches per injection, with each bunch containing an intensity of 1.6×10^{11} protons. Over the 147 days of operation in 2024, the bombardment of electrons further “conditioned” the beam screens, gradually reducing the number of electrons released (a bit like [scrubbing](#)). As a result, by the end of the run, the heat deposited in the beam screens was slightly lower than at the start, even though the bunch train configuration and bunch intensity remained unchanged.

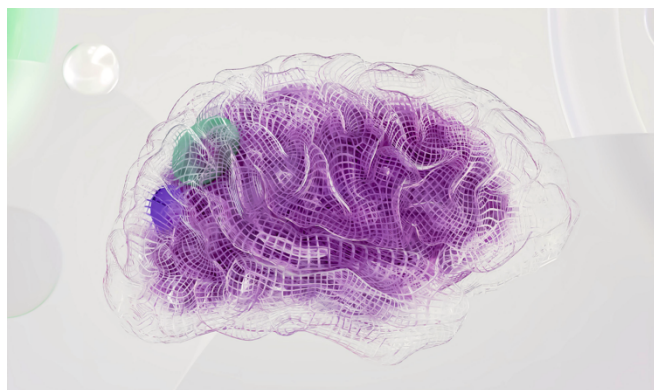


An overview of the heat load on the beam screens in each of the 8 arcs of the LHC. The limiting arc is 7-8, where the heat load is highest. In addition to the absolute heat load, the graph also shows a reduction of the heat load in all arcs, including arc 7-8. (Image: CERN)

This reduction in heat deposition creates a small margin to adjust the bunch train pattern (e.g. 4 x 36 bunches or perhaps even 5 x 36 bunches) and/or slightly increase the bunch intensity (e.g. 1.7×10^{11} protons per bunch or even 1.8×10^{11} protons per bunch). These modifications would allow a modest increase in luminosity production, making the target of 120 fb^{-1} for 2025 a realistic goal.

AI treatments for stroke survivors

CERN partners with public and private organisations to improve stroke prevention, diagnosis and treatment



(Image: UMBRELLA project)

Each year, more than a million acute stroke cases occur in Europe, with nearly 10 million survivors facing long-term consequences. Data on strokes is plentiful but fragmented, making it difficult to exploit in data-driven treatment strategies. But the toolbox of the high-energy physicist is well adapted to this task. This is why CERN has joined more than 20 public and private partners in the [UMBRELLA](#) project, which aims to transform

stroke care across Europe. This five-year project is supported by the [EU's Innovative Health Initiative](#) and launched in October 2024.

UMBRELLA plans to use advanced digital technologies to develop AI-powered predictive models for stroke management. By standardising real-world stroke data and using tools such as imaging technologies, wearable devices and virtual rehabilitation platforms, UMBRELLA aims to refine every stage of care – from diagnosis to recovery. Based on post-stroke data, AI-driven insights will help clinicians to uncover root causes of strokes, improve treatment precision and predict patient outcomes, reshaping how stroke care is delivered.

Underpinning this endeavour is UMBRELLA's “U-platform” – a data ecosystem based on CERN's machine-learning platform, [CAFEIN](#). CAFEIN was initiated thanks to seed funding from CERN's [medical applications](#) budget and uses a decentralised or “[federated](#)” approach to train

machine-learning algorithms without exchanging data. The U-platform will enable secure, privacy-preserving training of advanced AI algorithms for personalised stroke diagnostics, risk prediction and treatment decisions without sharing sensitive patient data between institutions. Building on CERN's expertise, including its success in federated AI modelling for brain pathologies as part of the EU [TRUSTroke](#) project, the CAFEIN team is poised to handle the increasing complexity and scale of data sets required by UMBRELLA.

This ambitious initiative not only showcases CERN's role in pioneering federated-learning technologies but also underscores the broader

societal benefits brought by basic science. By pushing technologies beyond the state-of-the-art, CERN and other particle physics laboratories have fuelled innovations that have an impact on our everyday lives. The outcomes of the UMBRELLA project hold the potential to redefine stroke care, delivering life-saving advancements to millions.

This is a shorter summary of [this article](#) published by the CERN Courier.

Amedeo Habsburg & Luigi Serio

Clocking nature's heaviest elementary particle

The CMS collaboration has tested whether top quarks play by Einstein's rules all day and all night

In the first [study](#) of its kind at the Large Hadron Collider ([LHC](#)), the [CMS](#) collaboration has tested whether top quarks adhere to Einstein's special theory of relativity.

Along with quantum mechanics, Einstein's special theory of relativity serves as the basis of the [Standard Model](#) of particle physics. At its heart is a concept called Lorentz symmetry: experimental results are independent of the orientation or the speed of the experiment with which they are taken.

Special relativity has stood the test of time. However, some theories, including particular models of string theory, predict that, at very high energies, special relativity will no longer work and experimental observations will depend on the orientation of the experiment in space-time. Remnants of such Lorentz symmetry breaking could be observable at lower energies, such as at the energies of the LHC, but despite previous efforts, they have not been found at the [LHC](#) or other colliders.

In its recent [study](#), the CMS collaboration searched for Lorentz symmetry breaking at the LHC using pairs of top quarks – the most massive elementary particles known. In this case, a dependence on the orientation of the experiment would mean that the rate at which top-quark pairs are produced in proton–proton collisions at the LHC would vary with time.

More precisely, since Earth rotates around its axis, the direction of the LHC proton beams and the average direction of top quarks produced in collisions at the centre of the CMS experiment also change depending on the time of the day. As a consequence, and if there is a preferential direction in space-time, the top-quark-pair production rate would vary with the time of the day. Hence, finding a deviation from a constant rate would amount to discovering a preferential direction in space-time.

The new CMS result, which is based on data from the second run of the LHC, agrees with a constant rate, meaning that Lorentz symmetry is not broken and Einstein's special relativity remains valid. The CMS researchers used the result to set limits on the magnitude of parameters that are predicted to be null when the symmetry holds. The limits obtained improve by up to a factor of 100 upon results from a [previous search](#) for Lorentz symmetry breaking at the former [Tevatron](#) accelerator.

The results pave the way for future searches for Lorentz symmetry breaking based on top-quark data from the [third run of the LHC](#). They also open the door to scrutiny of processes involving other heavy particles that can only be investigated at the LHC, such as the [Higgs boson](#) and the [W](#) and [Z](#) bosons.

CMS collaboration

At the World Economic Forum 2025

CERN Director-General Fabiola Gianotti speaks and performs with cellist Yo-Yo Ma at the 2025 World Economic Forum in Davos



CERN Director-General Fabiola Gianotti and world-renowned cellist Yo-Yo Ma perform a duet of “Le Cygne” by Camille Saint-Saëns at the World Economic Forum’s annual meeting in Davos, Switzerland. (Image: World Economic Forum)

In January 2025, CERN Director-General Fabiola Gianotti attended the World Economic Forum’s annual meeting in Davos, Switzerland. She joined world-renowned cellist Yo-Yo Ma on stage for a

discussion on the connections between science, nature and art, before accompanying him on the piano for a duet of “Le Cygne” by Camille Saint-Saëns.

During the event, Gianotti highlighted CERN’s successful model of global collaboration.

“Only by uniting brilliant minds from around the world can we hope to find answers to the most complex and profound questions,” she said. “Both physics and music are forms of exploration. Arts, science, philosophy – it’s all culture, they are what make us human.”

During her time as CERN Director-General, Gianotti has been invited to Davos on several occasions, with contributions ranging from participating in [debates](#) to [co-chairing](#). And this year’s contribution is not her first time collaborating with Ma. You can watch their CERN Science Gateway performance from December 2023 [here](#).

Strengthening fire and rescue collaboration

CERN and Geneva Airport join forces by signing a new collaboration agreement



CFRS and SSLIA teams in training at Geneva airport on 6 December 2024. (Image: Marc Nas/CERN)

In December 2024, the CERN Fire and Rescue Service (CFRS) and Geneva Airport’s Fire and Rescue Service (*Service de Sauvetage et de Lutte contre les Incendies Aéroportuaires – SSLIA*) signed a collaboration agreement to enhance their training and preparedness.

The collaboration enables both teams to train in realistic fire scenarios at each other’s facilities – the CFRS at the airport and the SSLIA at CERN’s Prévessin training facilities.

By sharing their expertise, the CFRS and SSLIA will gain new perspectives and enhance their knowledge and practical skills, thus building a strong relationship and being better prepared for both planned and unforeseen challenges.

The exchange fosters skill development through exposure to new environments and diverse training methods. It provides the firefighting

teams from both organisations with opportunities for joint training on techniques such as compartmental firefighting, fire behaviour analysis, search and rescue, and suppression techniques using simulators. The proximity of the two organisations to each other makes this partnership not just practical but also sustainable, with minimal travel needed.



Marc Nas, CFRS operations leader (right), and Lt-Col Yves Kreutzer, for SSLIA, shake hands to seal the agreement before starting a real fire exercise. (Image: Marc Nas/CERN)

This new agreement builds on existing frameworks in place, notably a tripartite agreement with the [Service départemental d'incendie et de secours](#) (SDIS) in France and the [Service d'incendie et de secours](#) in Switzerland for firefighter training, and an agreement with the [Hôpitaux universitaires de Genève](#) (HUG) for medical emergencies. These agreements strengthen the Organization's ties with the Host States in relation to fire and rescue efforts and generate increased regional visibility and recognition for the expertise of the CFRS. Another collaboration, with the local company [DSM-Firmenich](#), is also under discussion with a view to exchanging know-how and practices in the context of hazardous materials training.

HSE unit

Swiss artist Amanda E. Metzger selected for the Connect residency

Arts at CERN, in partnership with the Swiss Arts Council Pro Helvetia, announces the artist selected for the latest edition of Connect



Portrait of Amanda E. Metzger. (Image : Tim Oosterlynck)

Swiss artist Amanda E. Metzger will come to CERN for a two-month residency as the artist selected for the latest edition of Connect. Metzger's artistic practice centres on network theory, authorship and the complexities of unquantifiable data. Working across installations, videos, objects and photographs, she explores how memories are created, measured, and shared.

Supported by the Arts at CERN team, the artist will meet CERN scientists and visit the Laboratory with

them for two months. By engaging with cutting-edge scientific research, she will explore new artistic forms and develop *Reconstructing Reality*, a project that examines how collective identity and memories can be gathered, preserved and reimagined through methods inspired by experimental physics.

Drawing on the data collection, storage and lossy reconstruction processes of particle detectors, *Reconstructive Reality* will seek to reconstruct fragmented memories and build a decentralised, multi-bodied identity reflective of the artist's vision. The work will culminate in a web-based platform where viewers can navigate interconnected, reconstructed memories through an interplay of text, video, audio and graphics.

"Art, with its kaleidoscopic nature, inspires shifting perspectives on how we engage with science. The artists in our programme excel at fostering new possibilities, and I am proud to host Metzger in our residencies. Her work and residency proposal truly embody this transformative quality, with CERN

providing a remarkable setting for these ideas to unfold,” says Mónica Bello, head of Arts at CERN. Now in its eighth edition, the Connect collaboration framework between Arts at CERN and the Swiss Arts Council Pro Helvetia has established itself as a vital platform for artistic engagement with physics and the scientific community at CERN. The programme has hosted artists such as AATB, Johanna Bruckner, Robin Meier and Vimala Pons for residencies at CERN, and offered shared, dual residencies at CERN and partner scientific institutions in India, Chile and

South Africa to Rohini Devasher and Elisa Storelli, Shaliesh BR and Lou Masduraud, Marcela Moraga and Dominique Koch, and Kamil Hassim and Ian Purnell.

The Connect jury comprised Mónica Bello, curator and head of Arts at CERN; Giulia Bini, head of programme and curator of “Enter the Hyper-Scientific” at EPFL Lausanne; and Federica Martini, head of the CCC – Critical Curatorial Cybermedia master’s programme at HEAD (Haute école d’art et de design) in Geneva.

Computer Security: Don’t let your data walk away

Data confidentiality, while not paramount for CERN’s physics data, is still important when it comes to protecting sensitive contractual, financial, personal, radiation- and science-related documents and files. Just because CERN is “academic”, this does not and must not imply that data confidentiality is not an issue. On the contrary, it is our joint obligation to keep secret what is intended to be seen by just a few privileged eyes. Otherwise, lost data might become a media nightmare for CERN, or an operational risk, with both reputational damage and potential legal and contractual consequences. That’s why a dedicated [Data Governance Office](#) has been created. Before they ask, have you thought about how you handle your private and professional data? Do you let it walk away?

Or do you ensure the confidentiality of your private data on your portable devices (laptop, tablet, smartphone)? How do you store that data? How do you make sure it’s properly backed up? Of course, the first step is access protection – making sure that nobody but you (and your trusted peers) can access your portable device. Password protection. Fingerprint reading. FaceID. Whatever, as long as it is “strong” enough. The next step is local data encryption, if possible. Your Windows computer comes with “[BitLocker](#)” – just switch it on (and don’t lose its recovery code). Ditto for your MacBook – “[FileVault](#)” is your friend. For Linux, [Ubuntu](#) is one example of encryption available [during set-up](#).

Things get a bit more complicated when it comes to backing up using removable media, e.g. flash

drives, pluggable hard disks or your “network-attached storage” at home. Ideally, they come with an intrinsic encryption method already installed and ready for you to employ (once again, don’t forget the recovery code). If this is only for back-up, physical protection in a physical safe at home, under lock and key, might suffice. But if the purpose is sharing data with a third party, encryption of any confidential data is a must. Note that email communication is usually not encrypted and is therefore as unprotected as a snail-mail handwritten postcard. As an alternative to encryption, you can use platforms like Dropbox or Google Drive to upload your files and make them available for download via a shared URL (which you should communicate secretly). Of course, you need to trust those platforms’ security policies and their data privacy statements. [CERNBox](#) can be a valid alternative as long as your [personal usage is not excessive](#).

For professional data, of course, CERNBox or any other CERN-managed storage system is the best and [ONLY official way to share data](#) with your colleagues. The easiest approach is to upload and share by granting access to the relevant CERN computing accounts, always following the “principle of least privilege”. Only those who need to know should be able to access those professional documents. We will discuss this more in a forthcoming Bulletin article on data handling. And if you happen to use a CERN-owned computer (i.e. paid for via a CERN budget code*), it is imperative that this device has local storage encryption enabled. The Computer Security Office

will enforce this in 2025 following a recommendation of the [2023 audit on CERN computer security](#). In addition, an upcoming [Subsidiary Policy](#) will require that local USB storage devices not be used for any professional data classified as other than “public”. Last but not least, remember that sharing (unencrypted) documents via email in no way

preserves confidentiality. Just don’t do it unless you have encrypted your files first. Otherwise, your data might just walk away...

**... but not a “Team” account.*

Computer Security Office

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Administrative Circular No. 30 (Rev.5) – “Financial benefits on taking up appointment and on termination of contract”

A revised version of Administrative Circular No. 30 (Rev.5), entitled "Financial benefits on taking up appointment and on termination of contract", has been approved by the Director-General and is available via the following [link](#).

This version cancels and replaces Administrative Circular No. 30 (Rev.4), entitled "Financial benefits on taking up appointment and on termination of contract", dated October 2022.

The amendments made concern the following areas:

Arrival and departure travel

As of 1 January 2025, all members of the personnel entitled to the payment of arrival and departure travel expenses will receive a lump-sum payment. The aim of introducing lump sums for everyone is to reflect the evolution of how travel is organised: nowadays, most members of the personnel arrange their arrival and departure travel themselves. The use of lump sums will enhance flexibility for those recruited by providing them with transparency and clarity about the amount they will receive and will ensure equal treatment for all categories of personnel entitled to travel

reimbursement. Moreover, it will allow administrative simplification.

For employed members of the personnel, the real costs incurred may be reimbursed in exceptional circumstances, where the lump sum is not sufficient (§18). In such cases, approval must be requested from the Head of the HR department and the request must be submitted, if possible, at least 21 calendar days before travel.

Based on a comparison of the lump sums per country and the real-cost reimbursements paid over the last two years, it is expected that some 1% of travellers each year may wish to invoke this clause.

The lump sums and a brief explanation of how they are established will be published in the Admin e-guide.

Single-quotation removal procedure

A single-quotation procedure for removals has been introduced in the interests of administrative simplification. This new procedure provides the possibility for staff members to submit only one quotation, provided that it remains below a ceiling (which depends on the country of origin or

destination, as applicable, and on the family situation).

The ceilings are based on those of comparable organisations and will be reviewed every three years, or earlier if deemed necessary. Analysis of past removals shows that about 90% of removals are likely to fall within the one-quotation procedure.

The three-quotation procedure will remain in place to cover exceptional cases.

Reinstallation indemnity for staff members appointed before 2007

In order to benefit from the reinstallation indemnity, staff members appointed before 2007 must take up residence outside a 20 km radius of their former place of residence and of the duty station. In the previous version, the Circular only addressed the first criterion; this oversight is now corrected.

Recovery of payment of the installation indemnity in the event of resignation of staff members and graduates

As of 1 January 2025, staff members are required to reimburse the installation indemnity if they

resign within the first six months and on a pro-rata basis, according to the number of months of service completed. In the previous version of the circular, reimbursement of the full indemnity was required in case of resignation during the first year, irrespective of the number of months of service completed. The rationale behind this change is to acknowledge the additional costs linked to integration in the local area that cannot be recovered upon departure.

Graduates will no longer be required to pay back the installation indemnity upon resignation.

Textual clarifications

Furthermore, two textual clarifications have been made. The first concerns the definition of “*place of residence*” on taking up appointment (§ 3.1a) and the second makes claims for benefits subject to the provision of “*evidence*”, rather than “*supporting documents*” (§40b).

The revised circular has entered into force on **1 January 2025**.

HR department

Data Classification Policy

CERN has just released, for internal distribution only, the [Data Classification Policy](#) document, which establishes a framework for classifying professional data entrusted to or originating from the Organization. The objective of classification is to provide persons dealing with information with a concise indication of its level of security and access, thus allowing them to decide whether specific handling is required, such as appropriate technical measures and controls for protecting the data and for managing the life cycle of the data.

For this purpose, only information that requires specific handling needs to be explicitly categorised. The Policy applies to:

1. all professional data processed by or on behalf of CERN, in any form or format; and
2. all persons and services that process data at CERN or on behalf of CERN.

The [full document](#) in English is available on CDS. The French translation will be available soon.

Document Management Systems – Guidelines

Over the years, CERN’s workflows and activities have resulted in a variety of software solutions being used to manage, store and track digital records. Based on an analysis of the document

management systems (DMSs) currently used at CERN, the recently published Document Management Systems document aims to provide users with guidelines to help match their needs

with the features and functionalities of the DMSs supported by CERN. The list of DMSs included in the document was approved by the Steering Group on Externally Provided Software, Systems, Hardware and Services ([STEPS](#)) in October 2024. Any future requests that cannot be satisfied by the services listed must be reviewed against the existing capabilities and workflows, evaluated by

the STEPS Technical Committee and approved by STEPS before any additional document management systems are introduced.

The full document can be found at the following link:

<https://cds.cern.ch/record/2921130/files/Document%20Management%20Landscape%20and%20Guidelines%201.0.pdf>

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

29 Jan–2 Feb | Knowledge Sharing | Palexpo | Art Genève | EN

[Art & Science Co-Lab at Art Genève](#)

30 Jan 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Long Baseline Neutrino Experiments and the enabling role of the CERN Neutrino Platform: DUNE and ProtoDUNE](#)

30 Jan 17:00 | At CERN | Online | Mentoring@CERN | EN

[Mentoring@CERN Information Session](#)

31 Jan 14:00 | Knowledge Sharing | Online | Micro-Talks | FR

[#14 Analyse du langage non verbal : jusqu'où peut-on cacher la vérité ?](#)

3 Feb 16:30 | Knowledge Sharing | Science Gateway Auditorium B | CERN Colloquium | EN

[Click Chemistry, Excellence and Equality in Science](#)

5 Feb 18:00 | Knowledge Sharing | CERN Science Gateway | Public Events | EN

[2nd CERN Art and Science Summit](#)

11 Feb 20:00 | Knowledge Sharing | CERN Science Gateway | Public Events | FR

[De Temps en Temps](#) – part of CERN's [Quantum!](#) season

13 Feb 13:30 | Knowledge Sharing | 160/1-009 | IPPOG & WIT | EN

[Special Particle Physics Masterclass for CERN Personnel](#)

14 Feb 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[The Voice of Science – "Hollywood physics" by Professor Carsten Welsch](#)

Registration now open

10–14 Mar | Accelerators | Ferney-Voltaire | CERN Accelerator School | EN

[Basics of Accelerator Physics and Technology](#)

14 Mar 14:30 | Knowledge Sharing | Online | CERN Alumni | EN

[Moving Out of Academia to Data Science & Analytics \(2nd edition\)](#)

15–27 Jun | Accelerators | Bulgaria | CERN Accelerator School | EN

[Pushing the Limits: Intensity Limitations in Hadron Beams](#)

17–26 Jun | Experiments | Lithuania | ISOTDAQ | EN

[ISOTDAQ 2025 – International School of Trigger and Data Acquisition](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events [happening today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch.

New AI Companion features on CERN Zoom

Meeting summaries, topic tracking and more: the new AI Companion features are now available on CERN Zoom

As of **6 January 2025**, Zoom's **AI Companion** is officially available on CERN Zoom. This handy tool is designed to make your life easier, with features such as meeting summaries, topic highlights and ways to catch up on live discussions or recordings. You can find the full rundown of what it can do on the [Videoconference service website](#).

AI Companion is **disabled by default**. If you'd like to use it, you'll need to [switch it on from your Zoom profile](#).

While the AI Companion is here to help streamline your work, we **strongly recommend reviewing any AI-generated content** before sharing it. For details on staying secure and avoiding risks, check out the [security and privacy guide](#).

To help you get started, you can [watch the recording](#) of the Zoom **training session** for the CERN community that took place on 14 January.

If you have any questions, please contact the [CERN Videoconference service team](#).

Road safety at CERN: respect the rules, respect each other

Safety isn't just about our workspaces; it extends to how we conduct ourselves on the roads on and around the CERN site

Safety isn't just about our workspaces; it extends to how we conduct ourselves on the roads on and around the CERN site. Every day, CERN's entrances see heavy traffic from cars, bicycles and pedestrians and, unfortunately, frequent disregard for the highway code creates serious safety risks.

The entrances to the CERN site are particularly accident-prone zones. Infringements of the highway code are regularly witnessed at Meyrin site entrances A, B and E, where accidents and near misses due to failure to follow the traffic rules are frequently reported.

All road users must abide by the French and Swiss highway codes on site, just as they do outside CERN. This is particularly important [in the winter and in wet or icy conditions](#). Speed limits must be followed, and drivers should be mindful of footpaths and cycle paths. The general speed limit on the CERN sites is 50 km/h, reducing to 30 km/h in some areas. Speed cameras have been deployed

in the most accident-prone spots to educate road users.

We all have a duty to report accidents and near misses, as such reports help the Organization decide what further measures should be taken to make the road infrastructure safer. If you are involved in or witness a road traffic incident, please report it within 24 hours using the [EDH Incident Declaration form](#).

By following road safety rules, each of us contributes to a respectful, safe environment for all.

In the event of an accident or other emergency, call the CERN Fire and Rescue Service on +41 22 76 74444.

For more information about road safety at CERN, visit the dedicated road safety page: <https://hse.cern/safety-risks/road-safety>

HSE unit and SCE department

The “Meyrin–Prévessin Express” pilot shuttle service becomes official

In February 2024, the SCE, SY and BE departments [launched a pilot project](#) for a new shuttle service between the Meyrin and Prévessin sites – [the Meyrin–Prévessin Express](#). This reinforced version of [Circuit 5](#) increased shuttles during peak times, including over lunchtime. After one year of service, the results and feedback are very positive. The Meyrin–Prévessin Express

has therefore now been made part of the official CERN shuttle service.

Furthermore, to extend its efforts to develop and adopt sustainable and low-carbon solutions for the campus, the SCE department will replace the whole shuttle fleet by 100% electric vehicles from 1 February.

Preventing winter falls and injuries

Some tips and advice from the CERN infirmary

Every winter brings an increase in the number of visits to the infirmary, particularly in connection with falls or injuries as a result of icy roads and paths or winter sports of one kind or another.

According to Suva, the Swiss national accident insurance fund, more than 34 000 employees are injured practising winter sports every year in Switzerland ([2024](#) figure); in France, that number reaches 122 000 (winter 2021–2022 statistics from the [French monitoring body for winter sport accidents](#)).

The most common injuries in downhill skiing (46%) affect the lower limbs (hips, legs, knees and feet), while 33% affect the upper limbs (shoulders, arms, elbows, hands and fingers), a figure that rises to 42% for snowboarding (see the latest statistics from [Suva](#), dated 2024).

Most of these injuries can be avoided by proper physical preparation and using the right kit. Here are a few basic tips for skiers and snowboarders:

- Prepare your body: strengthening your muscles and cardiovascular system will help to reduce the risk of an accident.
- Take care of your kit: your skiing and snowboarding equipment should be checked in order to avoid falling or twisting your knee because of badly adjusted bindings or boots.
- Protect your head: prevent the risk of head trauma by wearing a [helmet](#) that meets the safety standards and fits your head properly.
- Warm up and stretch: a key recommendation before hitting the slopes.
- Wear wrist guards if you snowboard.
- Eat regularly and drink hot drinks to stay hydrated.
- Match your speed to your abilities.

We hope these tips will help you stay safe this winter while you enjoy the great outdoors.

Medical Service

Obituaries

John Pett (1942 – 2024)



(Image: John's family)

It is with deep sadness that we share the news of the loss of our friend and colleague, John Pett, former member and leader of the LHC Power Converter group. He was an important member of the group, contributing with his great experience, knowledge and efforts in innovation.

John was born in December 1942 in Toddington, UK. His early interest in electronics led him to study engineering and to work at PYE Electronics in Cambridge. In 1966, a colleague saw a CERN job advert in the New Scientist and John applied. The recruitment was fast: the interview was in September, and he started work in October.

John began at CERN in the Beam Transfer (BT) group of the Intersecting Storage Rings (ISR). He developed a Digital–Analog Converter (DAC) for the control of all the BT power supplies, enabling the first computer control. He then went on to develop a high-precision 16-bit DAC for the ISR main ring supplies, which was later also deployed in the pulsed beam lines and the Antiproton Accumulator (AA). He also studied the new switch-

mode technology and designed the first 6 kW converters for some ISR correctors. Later, he was instrumental in creating a new control system for all ISR power supplies, based on CAMAC control and the first microprocessors.

In 1980, the group's focus shifted to the LEP project and John led the development of a new control system for the ~900 power converters, based entirely on microprocessors, a real-time operating system, hybrid DACs and Analog–Digital Converters (ADCs) and the 1553 field bus. He was also the divisional representative on CERN committees for training and standardisation of electronic components.

In autumn 1989, John was appointed group leader and managed the merger of the LEP and SPS converter groups and prepared for the forthcoming LHC project. He developed a groundbreaking 22-bit Delta-Sigma ADC over a number of years with unprecedented accuracy and speed. It would go on to be part of, again, an entirely new control system for the LHC converters. John successfully managed this development, based on digital signal processing, multi-microprocessors and the new real-time WorldFIP fieldbus.

In John's last years at CERN, up to his retirement in 2002, he also made significant design contributions to the current calibration system, which is the basis for the outstanding performance of the LHC power converters. These calibration devices have been widely recognised and some are used in national standards laboratories.

His achievements were presented in several papers to the Institute of Electrical and Electronics Engineers (IEEE) conferences and Transactions journal, thus gaining international recognition outside CERN.

After a few years of struggle with illness, John passed away on 12 December 2024. His contributions to high-precision electronics at CERN were outstanding. He was very persistent in attacking and solving difficult problems and was a good partner in discussions, not afraid to question statements and opinions. He also had a great

sense of British humour, and we laughed a lot. John will be remembered as a great professional engineer and a good friend.

His friends and colleagues

Ombud's corner

Understanding and overcoming “resistance to change”

As the new year begins and we look ahead to the changes that await us at the Laboratory, we may find ourselves feeling a mixture of enthusiasm and fear of the unknown. You may have heard the term “resistance to change”, which is usually defined as a natural human reaction to new routines, habits, workflows, processes and equipment in the workplace.

Managers often assume that resistance to change is inevitable and will gradually fade away once the changes are in place. Ron Carucci, a renowned speaker on leadership and organisational change, believes that resistance to change is a myth. What human beings fear is change that threatens to exclude them. In his view, it is this anxiety-generating fear of being sidelined that can compromise even the best managerial initiatives. I would add that any major change represents a loss and that it is important to recognise this. For example, you might lose a sense of efficiency when you move to a new role, or lose a strong collaborative relationship with a colleague after a restructure. Even desired changes where we are both actors and beneficiaries have an impact on us, requiring us to take time to adapt and expend a lot of effort.

Here are three key points to help you to overcome resistance to change, taking the human factor into account.

1. **Analyse** the human impact of the change: what consequences will it have on the activities of the team members, how their work is organised and their motivation?
2. **Announce** changes clearly, early enough to allow time to reflect and to consult everyone who will be affected.

3. **Manage** the change, which requires listening, sharing and communicating. By making themselves available to the members of their teams, managers can avoid unfounded concerns and give everyone the time and tools they need to get used to the new situation.

Change shakes up our habits and well-established relationships, so we need to be well prepared and take the time to establish new points of reference and collaborations.

If you would like to talk about this subject or understand how the points above can be applied to a particular situation, please contact me. As Ombud, my main purpose is to provide a non-official channel for conflict resolution, as well as to give impartial help and advice. My role is also to offer my opinion on the implementation of the Code of Conduct.

Thank you for your attention, and I look forward to seeing you soon.

Marie-Luce Falipou

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>.

The Ombud is available from Monday to Friday in office 500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at ombuds@cern.ch.

More information can be found on the Ombud's website: <https://ombuds.web.cern.ch/>